Immune Deficiency and Related Diseases

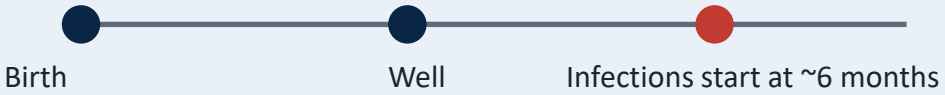
Year 2 Medicine · Clinical Immunology

By the end you should be able to…

1. Define primary vs secondary immunodeficiency
2. Explain what B cells, T cells, phagocytes and complement do
3. Classify immunodeficiency by affected component
4. Link characteristic infections to the defect
5. Describe key disorders in each group
6. Outline first-line investigations
7. Summarise prevention and treatment
8. Recognise dysregulation, autoimmunity and malignancy

A child who keeps getting sick

- Boy, 8 months old
- Third bacterial chest infection and second ear infection since about 6 months
- Well and thriving in the first months
- Immunoglobulins very low; B cells markedly reduced



QUESTION FOR THE ROOM

Why do some patients get infections more often, or more severely, than others?

Answer on slide 8

Source: Davidson p. 71 (antibody deficiencies present in infancy once maternal immunoglobulin has waned). Case written for teaching.

The immune system in one picture

INNATE: fast, built in

Phagocytes

Engulf and kill microbes

Main enemy

Bacteria, fungi

Complement

Tag, inflame, lyse

Main enemy

Neisseria, encapsulated bacteria

ADAPTIVE: specific, learned

B cells

Make antibodies (Ig)

Main enemy

Encapsulated bacteria

T cells

Kill infected cells; help B cells

Main enemy

Viruses, fungi, protozoa

Key idea: Defects in one arm cause infections typical of that arm

Source: Foundational physiology; infection patterns from Davidson pp. 70-73.

What is immunodeficiency?

Part of the immune system is missing or fails to work properly

Consequences

- **Recurrent infection**
- **Autoimmunity**
- **Autoinflammatory disease**
- **Allergy**
- **Bone marrow failure**
- **Malignancy (EBV-driven)**

Primary (inborn)

- Inborn errors of immunity: over 550 single-gene disorders
- Many present in childhood, not all

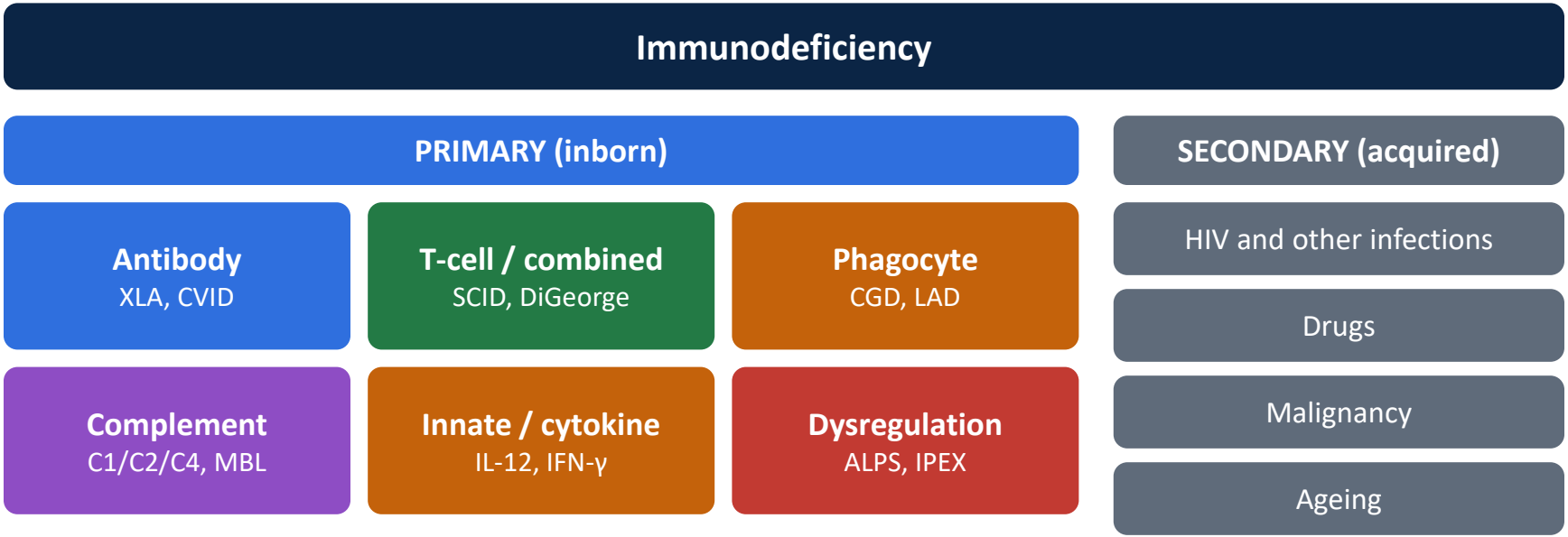
Secondary (acquired)

- External causes: infection, drugs, malignancy, ageing
- Much more common

Clinical takeaway: Infection is only one face of immune deficiency; also think autoimmunity and malignancy

Source: Davidson p. 70; p. 72 (CVID adult-onset); p. 73 (T-cell disorders can present in adults).

Classifying immunodeficiency



Same colours are used for every component through the lecture

Source: Davidson p. 70 and Box 4.9 (adapted from the IUIS 2022 classification).

Which component, which problem?

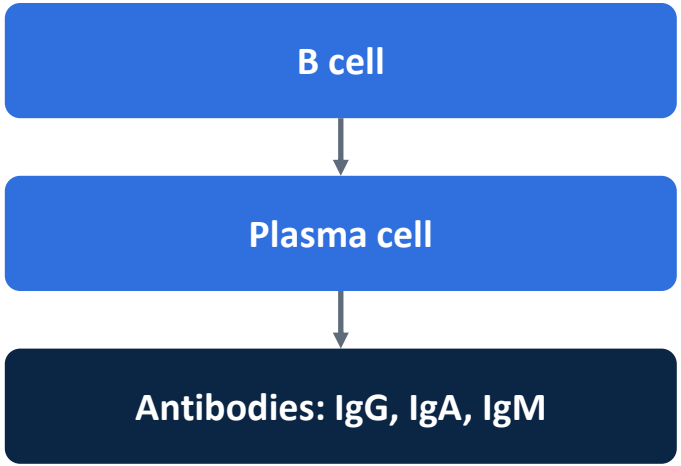
Component	Normal role	When defective
B cells / antibody	Make antibodies	Encapsulated bacteria; airway and gut
T cells	Cellular immunity; help B cells	Viral, fungal, protozoal infections
Phagocytes	Migrate, engulf, kill	Bacterial and fungal infections; abscesses
Complement	Tag, inflame, lyse	Neisseria; autoimmunity (classical pathway)
Innate / cytokine	Activate macrophages; sense microbes	Mycobacteria; HSV encephalitis; candidiasis
Regulation	Switch off responses; remove excess cells	Autoimmunity, lymphoproliferation, cytopenias

Clinical takeaway: Ask “what is the organism?” and the failing component usually follows

Source: Davidson pp. 70-73 and Box 4.9. Complement function: standard physiology.

Antibodies and B cells

- B cells mature into antibody-secreting cells; antibodies are immunoglobulins (Ig): IgG, IgA, IgM
- Primary antibody deficiency = abnormal B-cell function
- Recurrent bacterial infections of the respiratory and gastrointestinal tract
- Typical organisms: Streptococcus pneumoniae, Haemophilus influenzae (encapsulated)
- Usually presents in infancy, once maternal IgG has waned



Infancy: maternal IgG wanes, then infections begin

Source: Davidson p. 71; Fig. 4.9 as conceptual guide.

Selective IgA deficiency

1 in 600

northern Europeans; most common primary antibody deficiency

~30%

have recurrent mild respiratory and gut infections

< 0.05 g/L

IgA low or undetectable

- IgG and IgM normal (IgG sometimes raised)
- Usually asymptomatic, often found incidentally
- No specific treatment generally needed

Clinical takeaway: Common and often silent: treat the symptoms, not the number

Source: Davidson p. 72 and Box 4.10.

Common variable immune deficiency

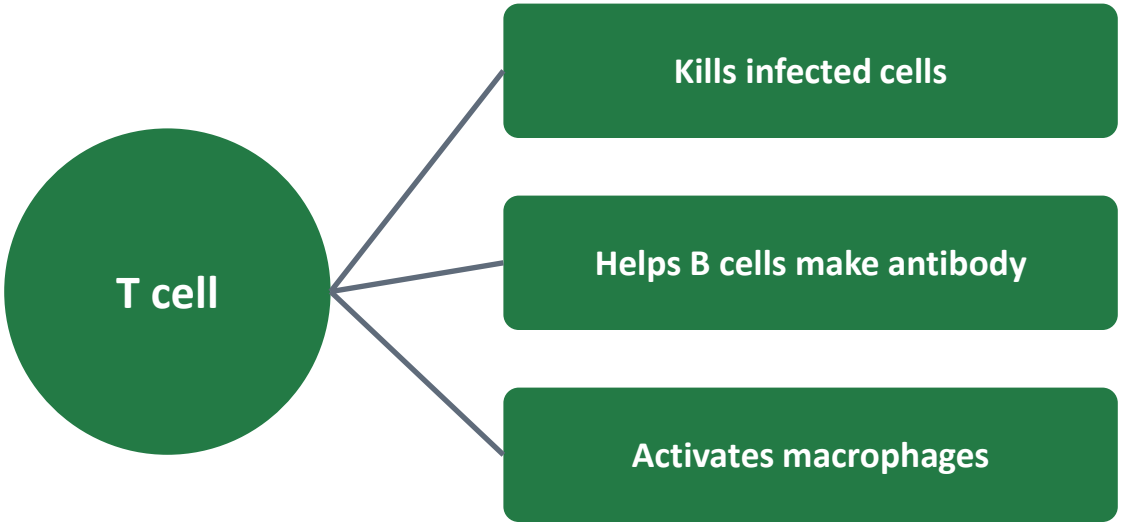
CVID: low IgG and IgA (± IgM)
No antibody response · B cells normal or low, not absent
Heterogeneous · usually adult-onset · cause usually unknown

Infection	Autoimmunity	Malignancy
Recurrent infections; bronchiectasis	Immune thrombocytopenia; autoimmune haemolytic anaemia	Lymphoproliferative disease

Clinical takeaway: Adult with recurrent chest infections plus autoimmune cytopenia: think CVID

Source: Davidson p. 72 and Box 4.9.

Why T cells matter



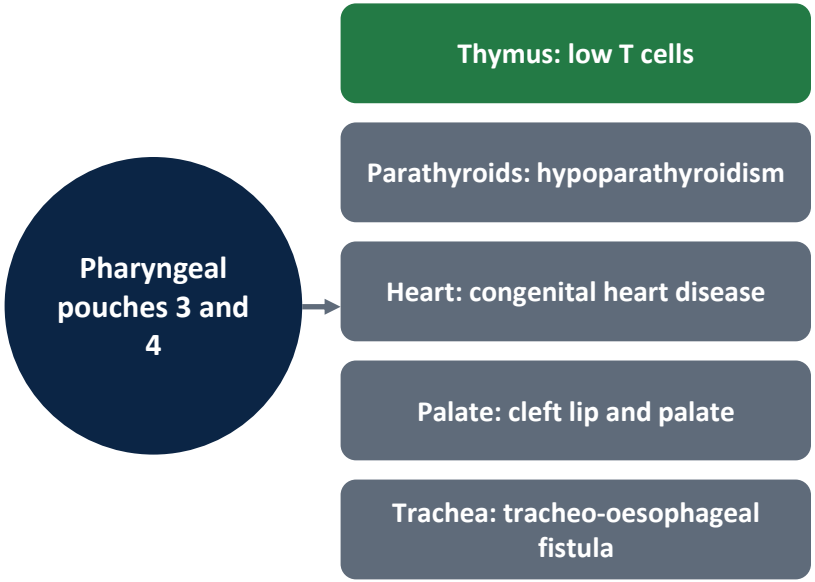
If T cells fail

- Viral, protozoal and fungal infections
- Antibody production often poor too
- Usually childhood, but adults can present

Source: Davidson p. 73; Fig. 4.9 legend. T-cell roles: foundational physiology.

DiGeorge syndrome

- Third and fourth pharyngeal pouches fail to develop; usually a 22q11 deletion (autosomal dominant)
- Thymus absent or poorly formed
- Very low circulating T cells despite normal bone marrow development
- Immune deficiency is very heterogeneous



Clinical takeaway: An infant with a heart defect and hypoparathyroidism may also have a T-cell problem

Source: Davidson pp. 73-74 and Box 4.9.

How a phagocyte kills a microbe

1 Recruit

Infection signals call neutrophils from the blood

2 Migrate

LFA1 on the neutrophil binds ICAM1 on the vessel wall

3 Engulf

The microbe is taken into the phagocyte

4 Kill

NADPH oxidase makes free radicals; cytokines (IL-12, IL-23, IFN-γ) activate macrophages

Question 2: a child has a very high neutrophil count during infection, but the infected site has no pus. Which step is broken?

Source: Davidson p. 73, Fig. 4.10. Recruitment and engulfment: standard physiology.

Chronic granulomatous disease

- Variants in one of six NADPH oxidase genes: failure of oxidative killing
- Commonest: CYBB, X-linked, mainly males
- Catalase-positive organisms: S. aureus, Burkholderia cenocepacia, Aspergillus
- Deep abscesses (lung, nodes, soft tissue, bone, skin) and granulomas
- Treat: IV antibiotics, drainage, long-term antibacterial and antifungal prophylaxis



Engulfed but not killed: abscesses and granulomas

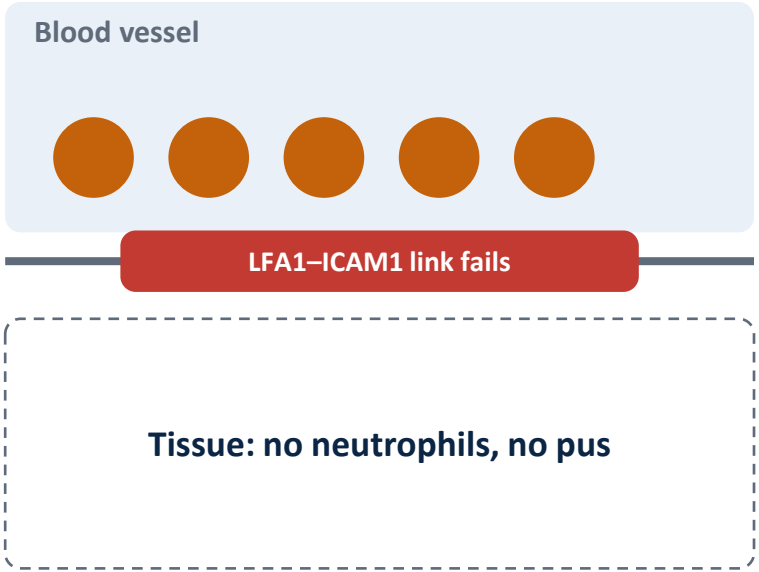
Clinical takeaway: Deep abscesses with S. aureus or Aspergillus, especially in a male: think CGD

Source: Davidson p. 70 and Box 4.9.

Leucocyte adhesion deficiency

- Adhesion molecules missing on leucocytes: they cannot leave the bloodstream
- Recurrent bacterial infections but no pus
- Blood neutrophil count may be very high in acute infection
- Tests: reduced or absent adhesion molecules on neutrophils

- LAD type 1: ITGB2 variants (beta-2 integrin, part of LFA1)



Clinical takeaway: Bacterial infection without pus and a high neutrophil count: suspect a migration defect

Source: Davidson p. 70, Fig. 4.10.

Cytokine and innate pathway defects

- Variants in genes for IFN-γ, IL-12, IL-23 or their receptors
- Macrophages are not activated: intracellular killing fails
- Particularly susceptible to mycobacterial infection
- One missing pathway, one characteristic infection

Defective pathway	Typical infection
IFN-γ / IL-12 / IL-23 axis	Mycobacteria (and Salmonella)
Toll-like receptor 3 (TLR3)	Herpes simplex encephalitis
IL-17 pathway (e.g. STAT1)	Mucocutaneous candidiasis

Clinical takeaway: Unusual mycobacterial disease in an otherwise healthy person: think IL-12/IFN-γ axis

Source: Davidson p. 70 and Box 4.9.

Complement: three clinical pictures

Complement tags microbes (opsonisation), drives inflammation and lyses bacteria (membrane attack complex, MAC)

Infection	Autoimmunity	Not infection
Encapsulated bacteria, especially Neisseria (MAC defence)	Classical pathway (C1, C2, C4) deficiency: risk of SLE	C1 inhibitor deficiency: recurrent angioedema

MBL (mannose-binding lectin) deficiency: about 5% of northern Europeans, many asymptomatic; more infection only with another immune insult (e.g. prematurity, chemotherapy); significance debated

Clinical takeaway: Think complement when Neisseria disease recurs or SLE is linked to classical-pathway deficiency

Source: Davidson p. 70. Complement function: standard physiology.

Complement: test and manage



C3 and C4 are widely available but may miss functional deficiency. CH50 = classical haemolytic pathway 50; AP50 = alternative pathway 50.

Management

Vaccinate

Meningococcal, pneumococcal, H. influenzae B

Prophylaxis

Lifelong penicillin against meningococcus

Be ready

Early assessment if infected; carry MedicAlert

Family

Screen at-risk relatives with functional assays

Clinical takeaway: Suspected complement defect: order CH50 and AP50, not just C3 and C4

Source: Davidson p. 71.

When immunity is also overactive

ALPS: autoimmune lymphoproliferative syndrome

IPEX: immune dysregulation, polyendocrinopathy, enteropathy

Malignancy: especially virus-driven (e.g. EBV); lymphoproliferative disease in CVID

Key idea: Infection susceptibility and immune dysregulation differ; one disease can cause both

Source: Davidson pp. 70, 72, 74 and Box 4.9.

Secondary immunodeficiency

Much more common than primary

External factors compromise the immune system. No inherited defect is needed.

Infection

HIV, measles

Drugs

Cytotoxic and immunosuppressive: transplant, autoimmunity, cancer

Malignancy

Disease itself compromises immunity

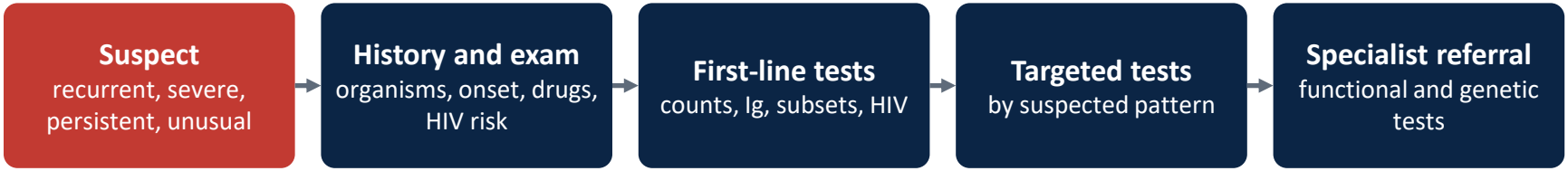
Ageing

Immune senescence; also extremes of life

Clinical takeaway: Before suspecting a rare inherited disorder, rule out HIV, drugs, malignancy and age

Source: Davidson pp. 70, 71, 74 and Box 4.11.

Approach to recurrent infection



First-line tests

- Blood count and absolute lymphocyte count (age-related ranges)
- Serum IgG, IgA, IgM, with electrophoresis to exclude secondary causes
- Lymphocyte subsets; HIV test if T-cell defect suspected

Pattern	Next step
Antibody	Specific IgG; killed vaccine, repeat at 6–8 weeks
T cell	Subsets; T-cell function tests
Phagocyte	Adhesion molecule and oxidative-killing tests
Complement	CH50, AP50, then components

Remember: Abnormal results need clinical context; no single test excludes immunodeficiency

Source: Davidson pp. 70-73 and Box 4.10. History and examination: beyond the excerpt.

Principles of treatment

Principle	Examples from the source
Treat infection promptly	Phagocyte defects: IV antibiotics, surgical drainage of abscesses
Prophylaxis in selected disorders	Antibacterial/antifungal (phagocyte); anti-Pneumocystis (T cell); penicillin (complement)
Immunoglobulin replacement	Antibody deficiency; IV or subcutaneous, lifelong; keep trough IgG in normal range
Vaccination	Complement: meningococcal, pneumococcal, Hib. Avoid live vaccines
Stem cell transplant and gene therapy	SCID: SCT first choice; gene therapy for ADA deficiency without a donor
Secondary immunodeficiency	Treat the underlying cause

Source: Davidson pp. 70-74, Box 4.11.

References

PRIMARY SOURCE

Davidson's Principles and Practice of Medicine

Clinical immunology, Immune deficiency, pp. 70–74

- Davidson, chapter 4 Clinical immunology: Immune deficiency, pp. 70–74; Box 4.9 (genetic causes), Box 4.10 (antibody deficiency investigation), Box 4.11 (immunoglobulin preparations), Fig. 4.9, Fig. 4.10
- Box 4.9 adapted in Davidson from Bousfiha A, et al. The 2022 update of IUIS Phenotypical Classification for Human Inborn Errors of Immunity. J Clin Immunol 2022;42:1508–1520
- No other external sources were used. Basic physiology (opsonisation, T-cell roles) is general teaching knowledge